FEU INSTITUTE OF TECHNOLOGY

COLLEGE OF COMPUTER STUDIES AND MULTIMEDIA ARTS

CCS0021L (INFORMATION MANAGEMENT)

Final Project

FINAL PROJECT - DATABASE SYSTEMS

<table>
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Introduction:
Jollibee Foods Corporation (JFC), as the largest fast-food chain in the Philippines with over 1,500 stores nationwide and numerous international branches, requires a sophisticated payroll system to manage its diverse workforce. The Jollibee Payroll System is designed to handle compensation for various employee types across multiple locations, ensuring accurate calculation of wages, deductions, and benefits while complying with Philippine labor laws and company policies.

Business Rules:
Based on research of Jollibee's operations and standard Philippine payroll practices, here are the key business rules:

A. Employee Classification & Branch Management

The system classifies staff to ensure correct pay structure and benefit entitlement while establishing the organizational hierarchy for reporting purposes.

I. **Branch Assignment:** Each employee is assigned to a specific Jollibee branch location.
II. **Employee Classifications:** Employees are tracked based on their employment status:
    - **Regular Full-Time:** Entitled to the complete benefits package.
    - **Probationary:** Under a six-month evaluation period; typically entitled to mandatory benefits like SSS and 13th-month pay.
    - **Part-Time/Casual:** Limited hours; benefit eligibility may be subject to minimum hour thresholds.
    - **Contractual:** Fixed-term employment.

B. Position-Based Pay Structure

Compensation components are defined by the employee's role, ensuring that pay is accurately calculated based on the nature of work performed.

a) **Crew Members:** Primarily compensated via hourly wages, plus mandatory service charge distribution.
b) **Cashiers:** Compensated via hourly wages, with system logic accommodating cash handling allowances.
c) **Cook/Kitchen Staff:** Compensated via hourly wages, with allowances for meals or, if applicable, hazard pay.
d) **Store Managers:** Paid a fixed monthly salary and are typically considered managerial employees, excluding them from service charge distribution.
e) **Assistant Managers:** Paid a fixed monthly salary with structured store performance incentives.

C. Shift and Attendance Rules

The system must precisely capture granular time inputs and automatically apply conditional premium pay calculations based on DOLE mandates.

- **Standard Shift:** Standard workday is eight (8) hours, with a minimum one-hour paid or unpaid break.
- **Overtime:** Work exceeding eight hours is compensated at a minimum of **125%** of the

standard hourly rate.

- **Holiday Pay:** Premium rates apply depending on the day type. For instance, work during a Special Non-Working Day is compensated at **130%** of the daily rate for the first eight hours.
- **Night Differential:** Mandatory premium of at least **10%** of the hourly rate for any work performed between **10:00 PM and 6:00 AM**.

D. Philippine Statutory Deductions

All compensation is subject to mandatory contributions, which must be deducted prior to the calculation of withholding tax.

- **SSS:** Social Security System contributions (Retirement and Social Benefits). The employee share is 5% of the Monthly Salary Credit (MSC) as of 2025.
- **PhilHealth:** Philippine Health Insurance Corporation premiums (Health Insurance).
- **Pag-IBIG:** Home Development Mutual Fund contributions (Housing and Savings).
- **Withholding Tax:** Income tax withheld by the employer, based on the progressive tax tables defined by the BIR (TRAIN Law) .

E. Company-Specific Benefits & Deductions

The system must track non-statutory pay components, particularly those affected by Philippine tax law's ceiling for non-taxable benefits.

- **Employee Discounts:** Benefits given, which fall under the category of *de minimis* benefits and are tax-exempt up to specific thresholds .
- **Meal Allowances:** Provided free or discounted meals per shift, tracked as a non-taxable *de minimis* benefit.
- **Uniform Deductions:** System allows for non-mandatory deductions, such as fees for lost or damaged uniforms.
- **Cash Shortage:** Allows for authorized deductions related to register discrepancies (requires appropriate documentation and authorization).

F. Payroll Schedule

The system must adhere to standard disbursement schedules and mandatory year-end pay rules.

- **Semi-monthly Payroll:** Standard pay schedule for employees, typically disbursed on the 15th and 30th or 31st of each month.
- **Cut-off Periods:** Time and attendance data is processed based on two corresponding periods: the 1st-15th and the 16th-end of the month.
- **13th Month Pay:** Mandatory benefit, equivalent to one-twelfth (1/12) of the total basic salary earned during the calendar year, calculated proportionally, and typically paid in December . This, along with other bonuses, is tax-exempt up to a cumulative limit of **₱90,000**.

Core Entities:

1. **BRANCH**

    - Attributes: Branch_ID, Branch_Name, Address, City, Store_Type (Mall, Standalone), Manager_ID
    - Relationships: Has many EMPLOYEES, Has many PAYROLL_RECORDS
2. **EMPLOYEE**
    - Attributes: Employee_ID, Branch_ID, Position_ID, First_Name, Last_Name, Hire_Date, Employment_Status, TIN, SSS_No, PhilHealth_No, PagIBIG_No
    - Relationships: Belongs to one BRANCH, Has one POSITION, Has many ATTENDANCE_RECORDS
3. **POSITION**
    - Attributes: Position_ID, Position_Title, Pay_Type, Base_Rate, Is_Managerial
    - Relationships: Held by many EMPLOYEES
4. **ATTENDANCE**
    - Attributes: Attendance_ID, Employee_ID, Shift_Date, Time_In, Time_Out, Regular_Hours, Overtime_Hours, Night_Hours
    - Relationships: Records time for one EMPLOYEE
5. **DEDUCTION_TYPE**
    - Attributes: Deduction_ID, Deduction_Name, Calculation_Type, Rate_Amount
    - Relationships: Used in many PAYROLL_DETAILS
6. **PAYROLL_TRANSACTION**
    - Attributes: Payroll_ID, Employee_ID, Pay_Period_Start, Pay_Period_End, Basic_Pay, Overtime_Pay, Holiday_Pay, Night_Differential, Gross_Pay, Total_Deductions, Net_Pay
    - Relationships: Created for one EMPLOYEE, Contains many PAYROLL_DETAILS
7. **PAYROLL_DETAILS**
    - Attributes: Detail_ID, Payroll_ID, Deduction_ID, Amount
    - Relationships: Part of one PAYROLL_TRANSACTION, References one DEDUCTION_TYPE

Key Relationships:

Relationship	Cardinality
One BRANCH employs many EMPLOYEES	1:N
One POSITION can be held by many EMPLOYEES	1:N

One EMPLOYEE has many ATTENDANCE records	1:N
One EMPLOYEE has many PAYROLL_TRANSACTIONs	1:N
One PAYROLL_TRANSACTION contains many PAYROLL_DETAILS	1:N
One DEDUCTION_TYPE appears in many PAYROLL_DETAILS	1:N

SQL:

Tables:

```
CREATE TABLE Standalone (
   standalone_id NUMBER PRIMARY KEY,
   branch_id NUMBER,
   manager_id VARCHAR2(10),
   branch_name VARCHAR2(50)
);
CREATE TABLE Mall (
   mall_id NUMBER PRIMARY KEY,
   branch_id NUMBER,
   manager_id VARCHAR2(10),
   branch_name VARCHAR2(50)
);
CREATE TABLE Branch (
   branch_id NUMBER PRIMARY KEY,
   branch_name VARCHAR2(50),
   address VARCHAR2(100),
   location VARCHAR2(100),
   store_type VARCHAR2(20),
   manager_id VARCHAR2(10),
   mall_id NUMBER,
   standalone_id NUMBER,
   CONSTRAINT fk_branch_mall FOREIGN KEY (mall_id) REFERENCES Mall(mall_id),
   CONSTRAINT fk_branch_standalone FOREIGN KEY (standalone_id) REFERENCES
Standalone(standalone_id)
);
CREATE TABLE Position (
   position_id NUMBER PRIMARY KEY,
   position_title VARCHAR2(50),
   pay_type VARCHAR2(20),
   base_pay_rate NUMBER(10,2)
);
CREATE TABLE Employees (
   employee_id NUMBER PRIMARY KEY,
   branch_id NUMBER,
   position_id NUMBER,
   first_name VARCHAR2(25),
   last_name VARCHAR2(50),
   hire_date DATE,
   employment_status VARCHAR2(20),
   TIN VARCHAR2(20),
   SSS_No VARCHAR2(20),
   PhilHealth_No VARCHAR2(20),
   Pagibig_No VARCHAR2(20),
   rate_amount NUMBER(10,2),
   CONSTRAINT fk_emp_branch FOREIGN KEY (branch_id) REFERENCES Branch(branch_id),
   CONSTRAINT fk_emp_position FOREIGN KEY (position_id) REFERENCES Position(position_id)
);
CREATE TABLE Attendance (
   attendance_id VARCHAR2(10) PRIMARY KEY,
   employee_id NUMBER,
```

```
    shift_date DATE,
    time_in TIMESTAMP,
    time_out TIMESTAMP,
    regular_hours VARCHAR2(50),
    overtime_hours VARCHAR2(50),
    night_hours VARCHAR2(50),
    CONSTRAINT fk_attendance_employee FOREIGN KEY (employee_id) REFERENCES
Employees(employee_id)
);
CREATE TABLE Deduction_Type (
    deduction_id NUMBER PRIMARY KEY,
    employee_id NUMBER,
    calculation_type VARCHAR2(20),
    rate_amount NUMBER(10,2),
    CONSTRAINT fk_deduct_employee FOREIGN KEY (employee_id) REFERENCES Employees(employee_id)
);
CREATE TABLE Payroll (
    payrollID NUMBER PRIMARY KEY,
    employeeID NUMBER,
    date_generated DATE,
    net_pay NUMBER(10,2)
);
CREATE TABLE Payroll_Transaction (
    payroll_id NUMBER PRIMARY KEY,
    employee_id NUMBER,
    pay_period_start DATE,
    pay_period_end DATE,
    basic_pay NUMBER(8,2),
    overtime_pay NUMBER(8,2),
    holiday_pay NUMBER(8,2),
    night_diff NUMBER(8,2),
    calculation_type VARCHAR2(20),
    gross_pay NUMBER(8,2),
    total_deductions NUMBER(8,2),
    net_pay NUMBER(8,2),
    branch_id NUMBER,
    CONSTRAINT fk_payroll_emp FOREIGN KEY (employee_id) REFERENCES Employees(employee_id),
    CONSTRAINT fk_payroll_branch FOREIGN KEY (branch_id) REFERENCES Branch(branch_id)
);
CREATE TABLE Payroll_Details (
    detailID NUMBER PRIMARY KEY,
    payrollID NUMBER,
    deduction_ID NUMBER,
    amount NUMBER(10,2),
    CONSTRAINT fk_details_payroll FOREIGN KEY (payrollID)
        REFERENCES payroll (payrollID),
    CONSTRAINT fk_details_deduction FOREIGN KEY (deduction_ID)
        REFERENCES deduction_type (deduction_ID)
);
```

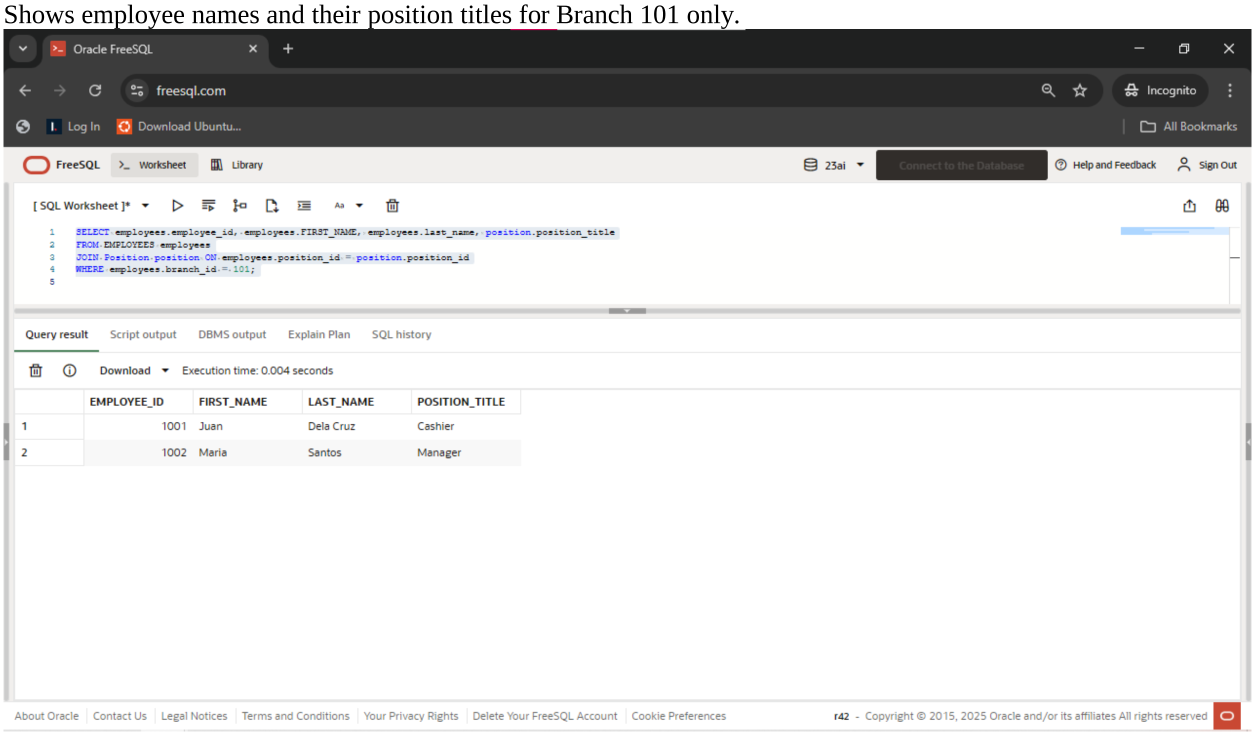
Sample Records:

```
INSERT INTO Standalone (standalone_id, branch_id, manager_id, branch_name)
```

```sql
VALUES (1, 101, 'M001', 'Standalone Branch A');

INSERT INTO Mall (mall_id, branch_id, manager_id, branch_name)
VALUES (1, 201, 'M002', 'Mall Branch A');

INSERT INTO Branch (
    branch_id, branch_name, address, location, store_type,
    manager_id, mall_id, standalone_id
)
VALUES (101, 'Branch A', 'Manila', 'NCR', 'Standalone', 'M001', NULL, 1);

INSERT INTO Branch (
    branch_id, branch_name, address, location, store_type,
    manager_id, mall_id, standalone_id
)
VALUES (102, 'Branch B', 'Quezon City', 'NCR', 'Mall', 'M002', 1, NULL);

INSERT INTO Position VALUES
(1, 'Cashier', 'Hourly', 150.00);

INSERT INTO Position VALUES
(2, 'Manager', 'Monthly', 30000.00);

INSERT INTO Employee (
    employee_id, branch_id, position_id, first_name, last_name,
    hire_date, employment_status, TIN, SSS_No, PhilHealth_No, Pagibig_No, rate_amount
)
VALUES (
    1001, 101, 1, 'Juan', 'Dela Cruz',
    DATE '2022-01-10', 'Active', '123-456', '55-2222',
    '44-1111', '33-3333', 150.00
);

INSERT INTO Employees (
    employee_id, branch_id, position_id, first_name, last_name,
    hire_date, employment_status, TIN, SSS_No, PhilHealth_No, Pagibig_No, rate_amount
)
VALUES (
    1002, 101, 2, 'Maria', 'Santos',
    DATE '2021-05-20', 'Active', '789-123', '66-7777',
    '55-5555', '44-4444', 30000.00
);

INSERT INTO Attendance VALUES
('A01', 1001, DATE '2024-11-20',
 TIMESTAMP '2024-11-20 08:00:00',
 TIMESTAMP '2024-11-20 17:00:00',
 '8', '2', '0');

INSERT INTO Deduction_Type VALUES
(1, 1001, 'Percentage', 0.05);

INSERT INTO Deduction_Type VALUES
```

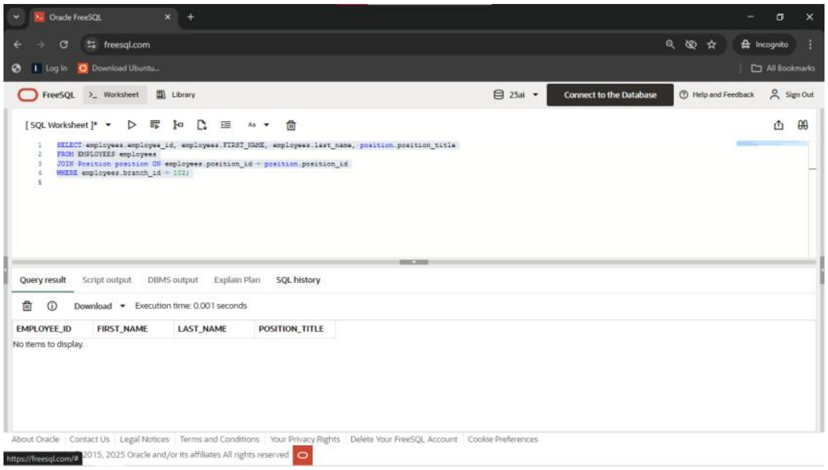
```
(2, 1002, 'Fixed', 500.00);

INSERT INTO Payroll_Transaction (
    payroll_id, employee_id, pay_period_start, pay_period_end,
    basic_pay, overtime_pay, holiday_pay, night_diff,
    calculation_type, gross_pay, total_deductions, net_pay, branch_id
)
VALUES (
    1, 1001, DATE '2024-11-01', DATE '2024-11-15',
    8000, 500, 0, 200,
    'auto', 8700, 300, 8400, 101
);

INSERT INTO Payroll_Details VALUES (1, 1, 1, 400);
INSERT INTO Payroll_Details VALUES (2, 1, 2, 300);
```

Query1:
Shows employee names and their position titles for Branch 101 only.

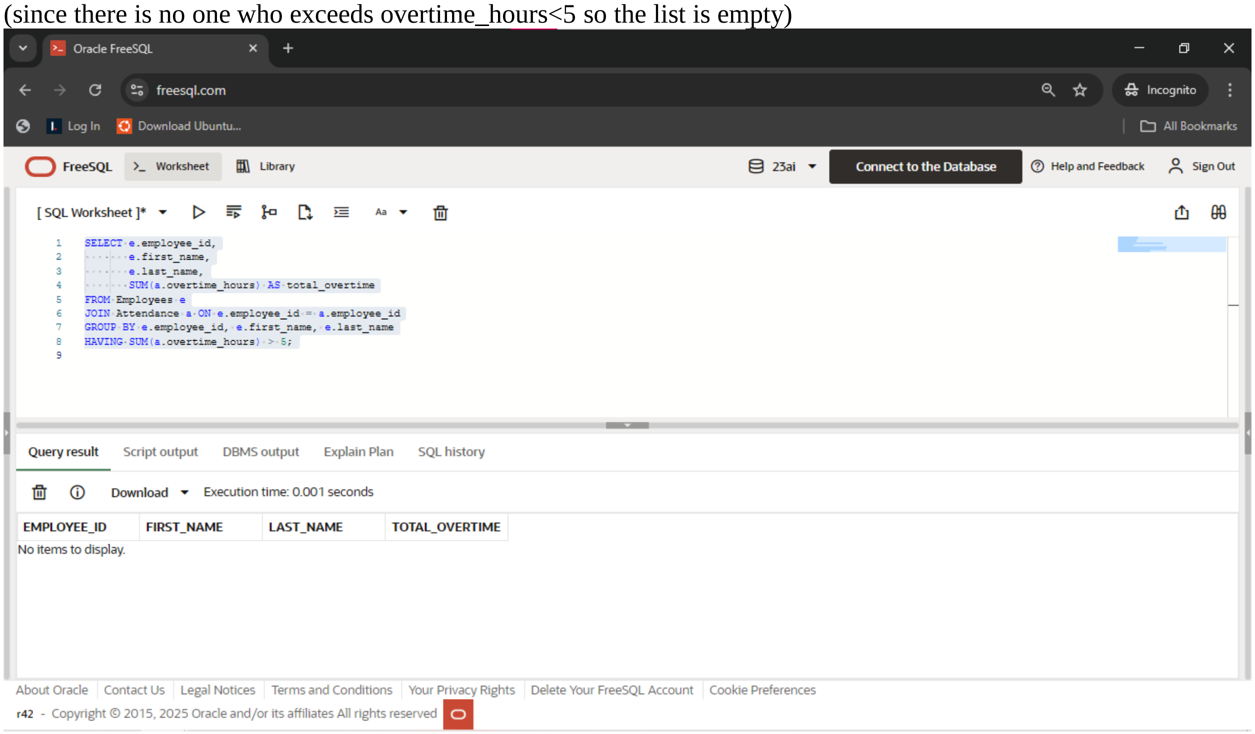



```
SELECT employees.employee_id, employees.FIRST_NAME, employees.last_name, position.position_title
FROM EMPLOYEES employees
JOIN Position position ON employees.position_id = position.position_id
WHERE employees.branch_id = 101;
```

(if in another branch)

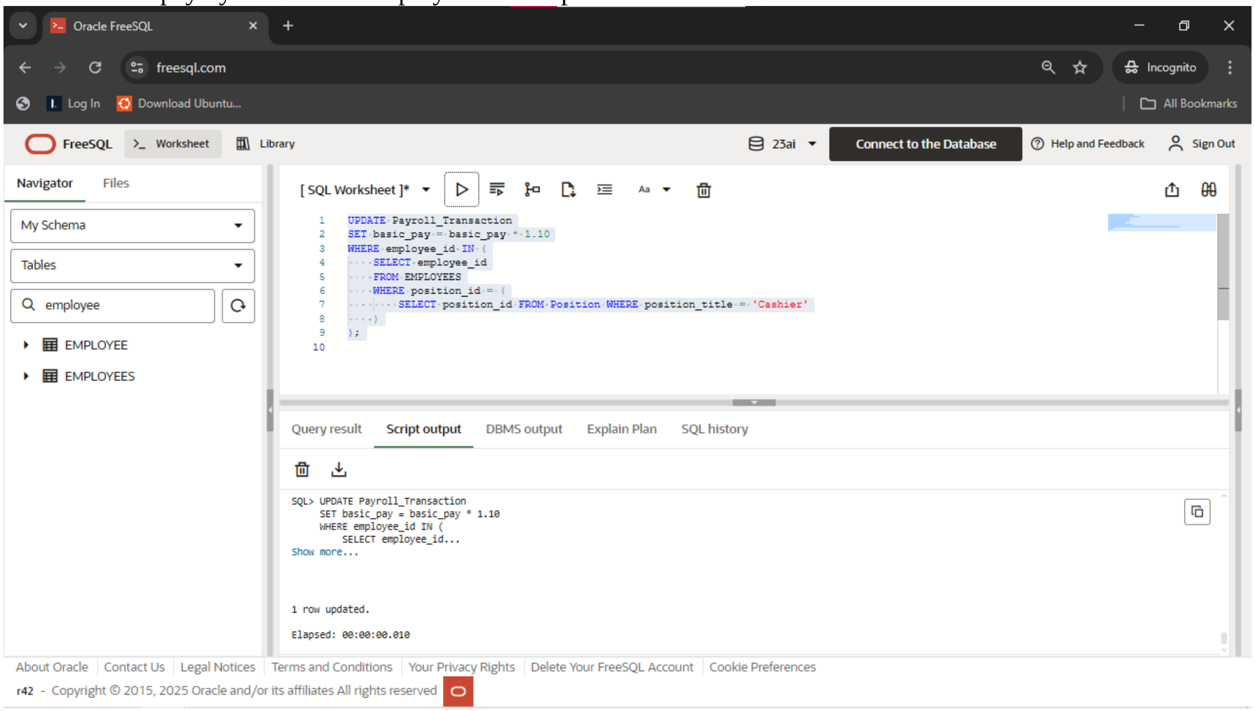


Query2:
Calculates total overtime hours per employee and shows only those with more than 5 hours.
(since there is no one who exceeds overtime_hours<5 so the list is empty)




```
SELECT e.employee_id,
    e.first_name,
    e.last_name,
    SUM(a.overtime_hours) AS total_overtime
FROM Employees e
JOIN Attendance a ON e.employee_id = a.employee_id
GROUP BY e.employee_id, e.first_name, e.last_name
HAVING SUM(a.overtime_hours) >
```

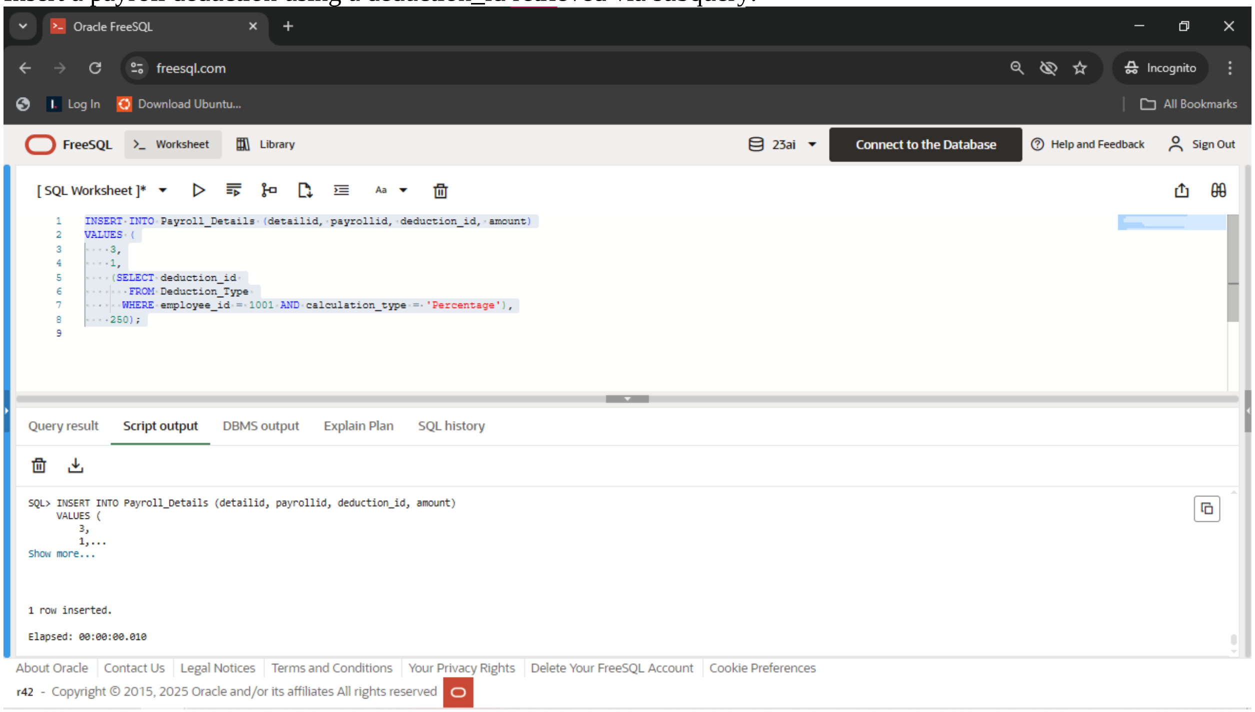
Query3:
Increase basic pay by 10% for all employees whose position is Cashier.




```
UPDATE Payroll_Transaction
SET basic_pay = basic_pay * 1.10
WHERE employee_id IN (
    SELECT employee_id
    FROM Employees
    WHERE position_id = (
        SELECT position_id FROM Position WHERE position_title = 'Cashier'
    )
```

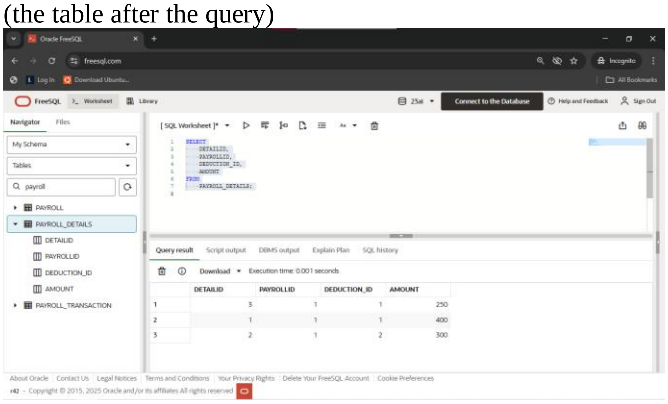
Query4:

Insert a payroll deduction using a deduction_id retrieved via subquery.

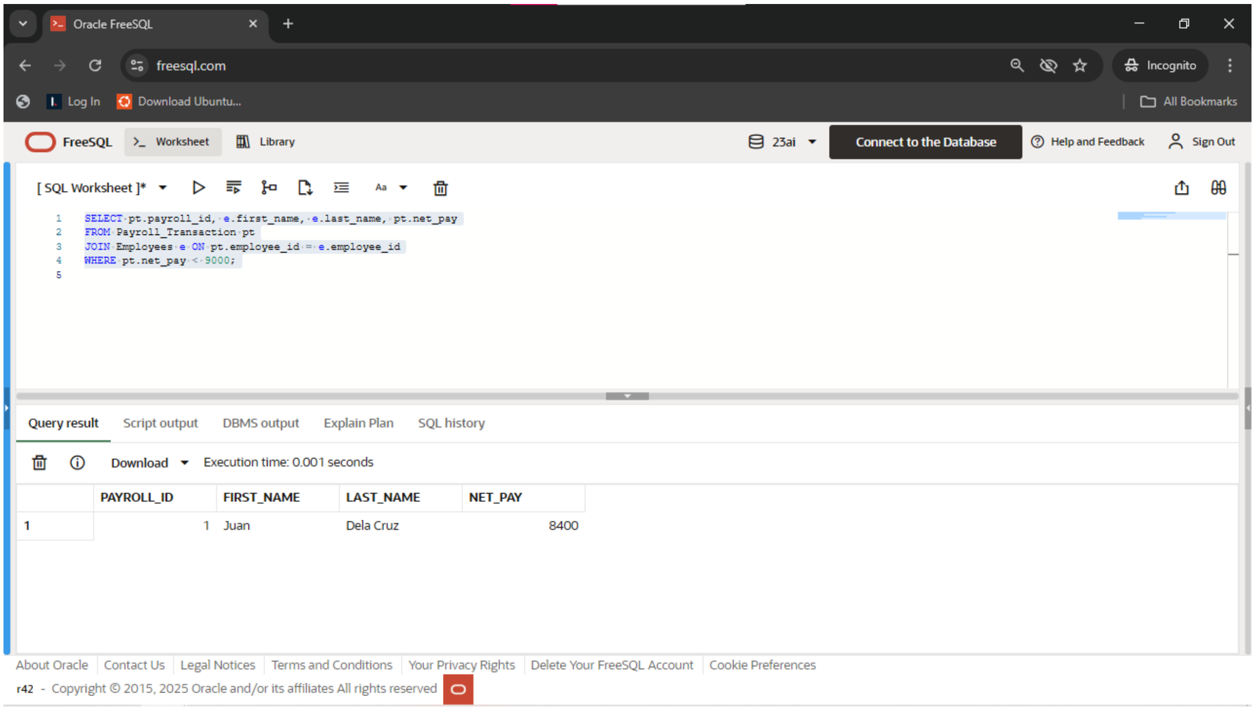


INSERT INTO Payroll_Details (detailid, payrollid, deduction_id, amount)
VALUES (
  3,
  1,
  (SELECT deduction_id
    FROM Deduction_Type
   WHERE employee_id = 1001 AND calculation_type = 'Percentage'),
  250);

(the table after the query)

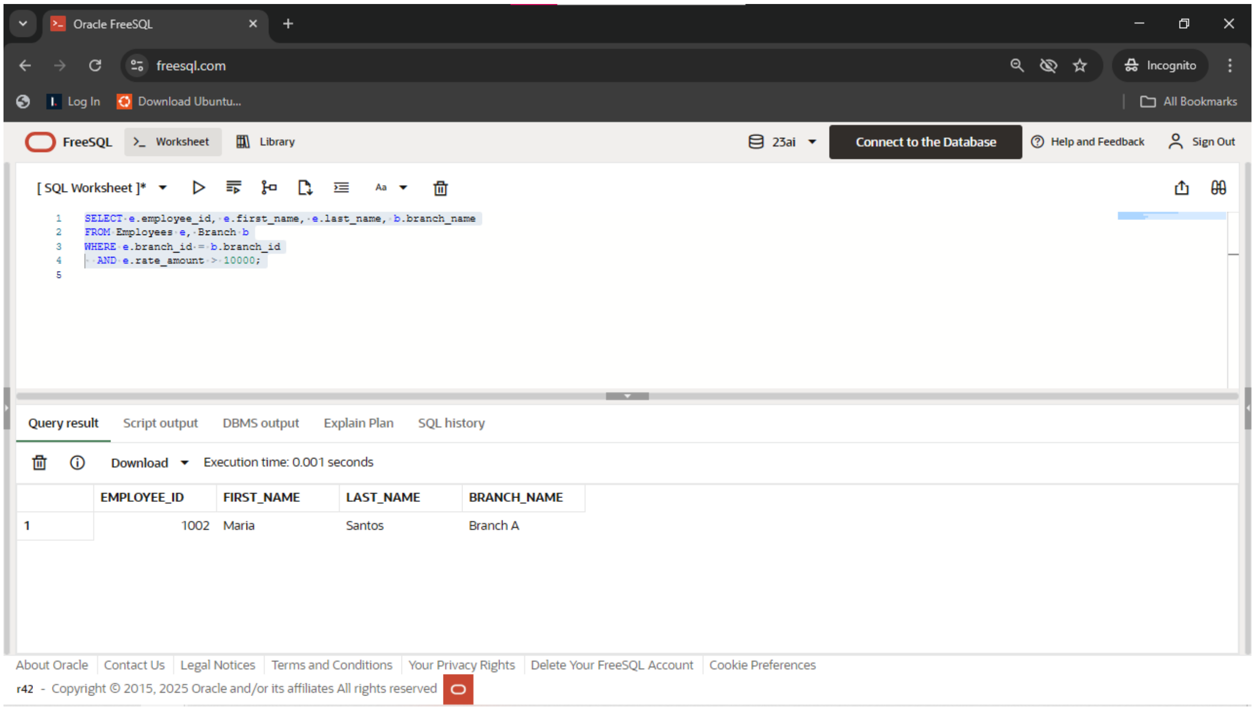


Query5:
Display payroll and employee info only for employees with net pay < 9000.




```
SELECT pt.payroll_id, e.first_name, e.last_name, pt.net_pay
FROM Payroll_Transaction pt
JOIN Employees e ON pt.employee_id = e.employee_id
WHERE pt.net_pay < 9000;
```

Query6:
Shows employee names and their branch names for employees earning more than 10000.




```
SELECT e.employee_id, e.first_name, e.last_name, b.branch_name
FROM Employees e, Branch b
WHERE e.branch_id = b.branch_id
  AND e.rate_amount > 10000;
```